NYLP AI HACKATHON 2026 · PROJECT DOCUMENTATION

Yaadon ki Baithak

یادوں کی بیٹھک

An AI agent that sits with your grandparents, listens to their stories in Urdu, and turns them into an illustrated family storybook, with their real voice inside.



Live app yaadon-ki-baithak.vercel.app

Repository github.com/CodeVoyager007/Yaadon-Ki-Baithak

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In ten chapters





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AI AGENTS	QUESTIONS ASKED	PAINTED PAGES	LANGUAGES	FACTS INVENTED



CHAPTER 01 / 10

The problem

کہانیاں جو کبھی لکھی نہیں گئیں

Every family has a storyteller. Most of their stories are never written down.

Every family has a Nani, a Dadi or a Nana with stories nobody wrote down: 1947 and the long journey, the old city, the wedding, the dishes that taste like home. They are told in Urdu, out loud, at bedtime, and when the storyteller is gone, the stories go with them.

Spoken, not written

Elders tell their stories out loud and in Urdu. Many are more comfortable speaking than typing, and some can't type at all.

A language gap

Grandchildren, often more fluent in English, can't follow the Urdu or the details everyone else took for granted.

Nobody has the time

Recording, transcribing, translating and illustrating a family's memories is weeks of work that never gets done.

Who it is for

Families in Pakistan and the diaspora who want to keep their elders' memories, and the elders themselves: people who would rather talk than type, in Urdu, and deserve to be listened to without interruption.

> *"Every elder is a library. When one goes, a whole library burns." We want to keep the books.*

The challenge we answered

The hackathon asked: *how might we use AI to solve a real-world problem and create a useful, accessible and responsible product?* Yaadon ki Baithak (*a gathering of memories*) is our answer for digital literacy and community: AI that listens patiently, asks only what matters, and turns a conversation into a book a family can keep for generations.

The solution

بات چیت سے کتاب تک

A conversation goes in. An illustrated storybook comes out.

1 Hudhud listens

Hudhud, a hoopoe from Urdu folk tales, asks the elder for consent on the record, then lets them tell the whole story in one go. It never interrupts.

2 Hudhud asks only what's missing

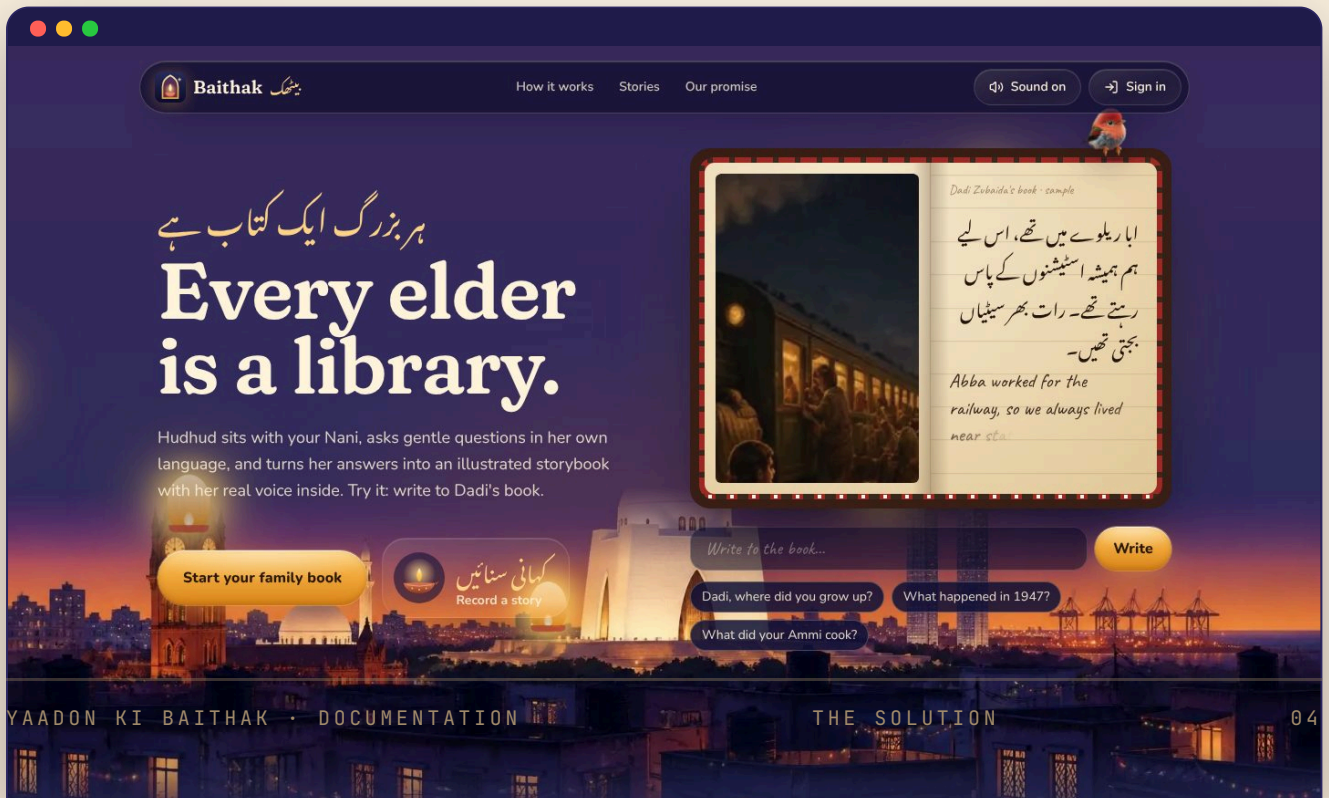
It reads the story and asks at most three gentle questions, spoken aloud in Urdu, about what a grandchild couldn't follow: who "he" was, where it happened, when. It notices grief or tiredness and slows down or stops.

3 A line of AI agents makes the book

The story writer edits it in the elder's own voice, the page setter lays out the pages, the art director plans a picture for every page, and the illustrator paints them in one storybook style.

4 The family reads it

It opens as a hardcover book with pages you turn by hand, in Urdu, English or both. One button plays the elder telling the whole story in their real voice. Download it as a PDF, share a private link, and keep every storyteller's books on one shelf.





CHAPTER 03 / 10

What it does

خصوصیات

Everything a family needs to keep a storyteller's voice.

Voice-first, in Urdu

The elder only talks. Hudhud's questions are spoken aloud in Urdu with English underneath.

Story first, questions after

The elder tells it all; at most three clarifying questions, only about real gaps.

Consent on the record

The elder agrees out loud before anything is saved, and the consent clip is kept.

An illustrated storybook

Four to eight pages, every page painted, in Urdu and English.

A real book to read

A hardcover that opens and pages that turn by dragging or with the arrow keys.

Their real voice

One player plays the whole story as it was told. No voice cloning, ever.

PDF download

One book or the whole family collection, with properly shaped Nastaliq.

Private share links

Read-only, unguessable links for relatives, ready for WhatsApp.

Many books, many tellers

Every storyteller's books on one family shelf, restored on any device after sign-in.

Never stalls

If a model is busy, the book is still made from the elder's own words and can be rewritten later.

Sample stories

Six bedtime tales open as full storybooks for anyone, no sign-in needed.

Sign in with Google or email

Supabase auth with private storage for voices and paintings.

CHAPTER 04 / 10

Inside the app

ایپ کے اندر

Real screens from the live product.

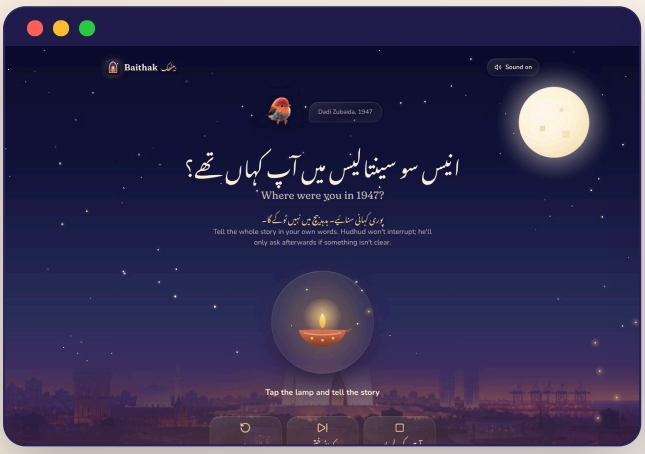


FIG 02 *Recording: the elder taps the lamp and tells the whole story.*

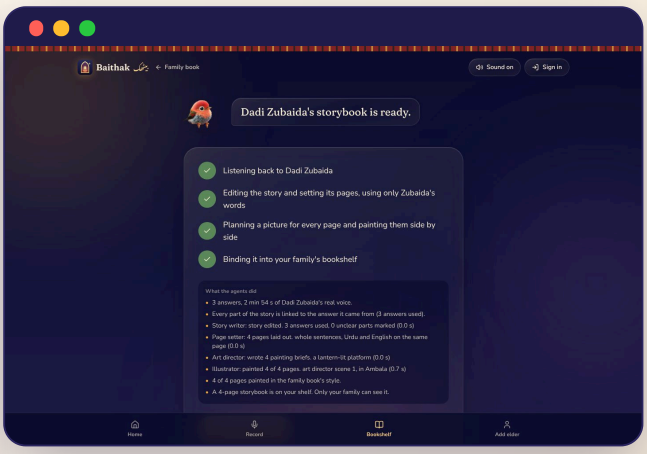


FIG 03 *The agents at work, each one reporting what it did.*



FIG 04 *The family bookshelf: storytellers, books, sharing and PDF.*

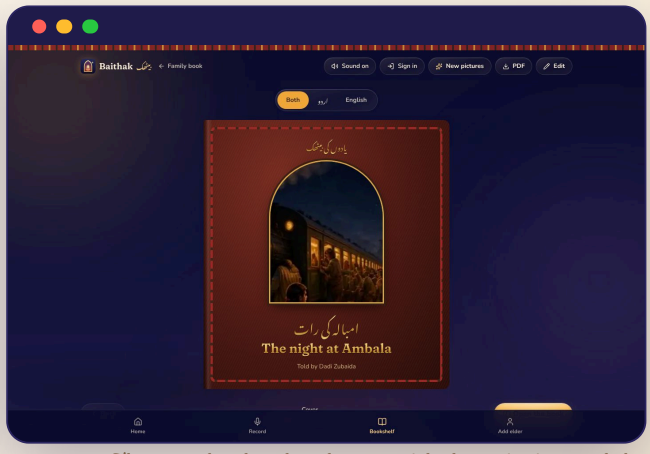


FIG 05 *The storybook: a hardcover with the painting and the title in gold.*

Reading the book

The reader is a hand-built 3D hardcover: the cover swings open, pages turn with a drag or the arrow keys, and Urdu, English or both can be shown. Each page has its painting beside the words.

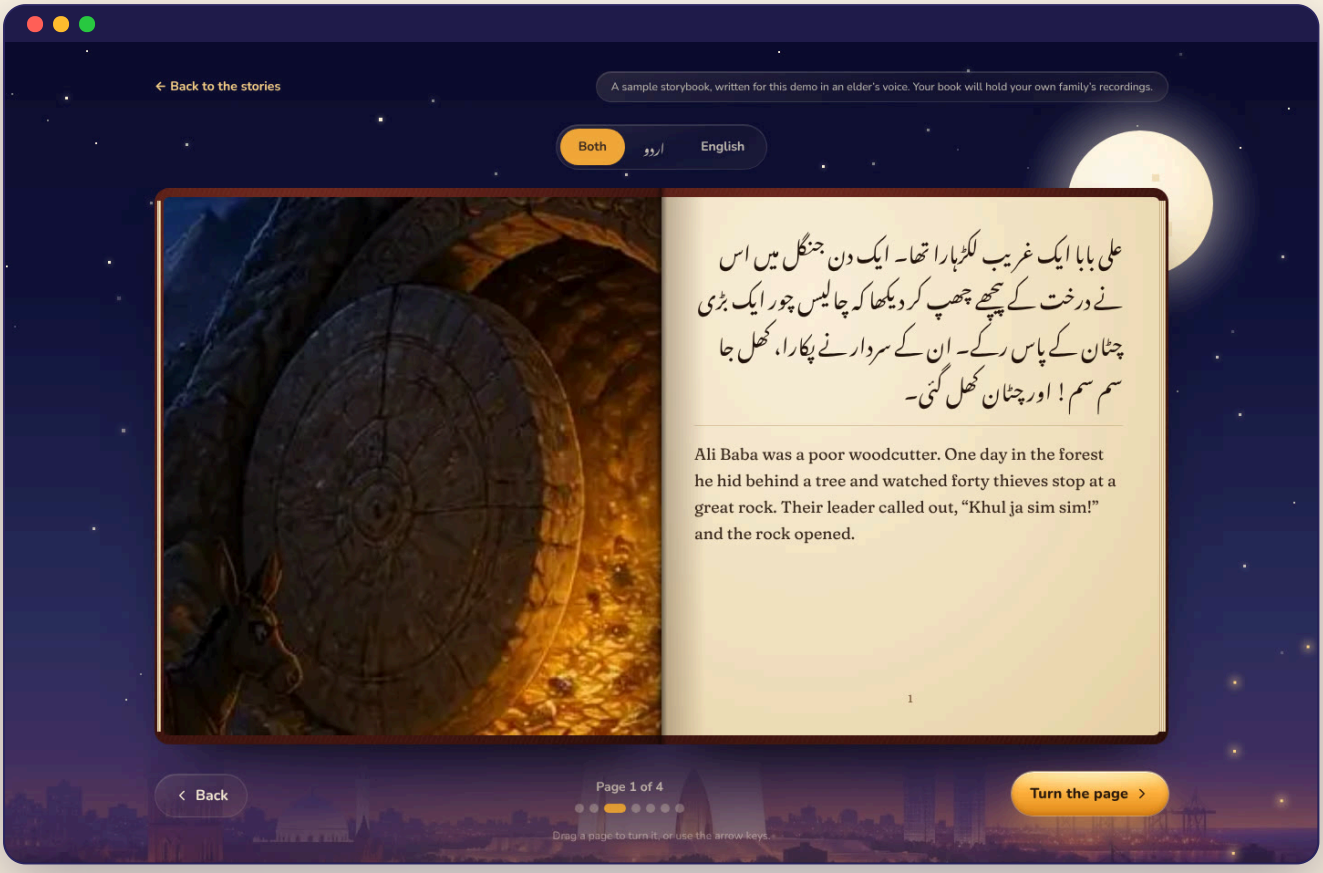


FIG 06 *An open spread from a sample storybook: the painting on the left, Urdu and English on the right.*

The sample bookshelf

Six bedtime tales (Umro Ayyar, Ali Baba, Gul Bakawali, Hatim Tai, Sheikh Chilli, the sparrow and the crow) open as full four-page storybooks for anyone, so visitors see the finished product before recording a word.

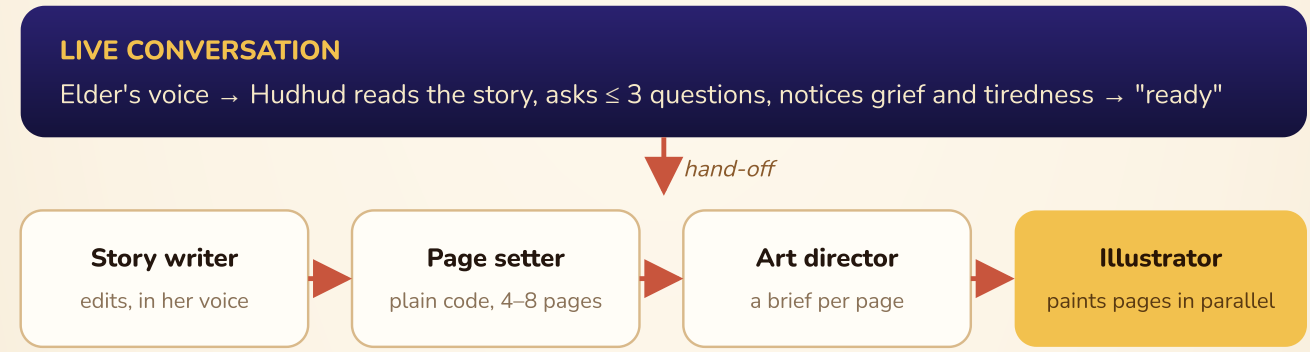




The agents

ایجنٹوں کی ٹیم

AI isn't decoration here: four agents make decisions, and code checks each one.



The orchestrator (plain code) runs the line, gives every step a fallback, and records a trace the family can see.

AGENT	WHAT IT DECIDES	RUNS ON
Hudhud (interviewer)	Every turn: ask a clarifying question, or hand off to the writer.	Gemini 2.5 Flash
Story writer	The edited story in the elder's voice: Urdu plus a faithful English translation, as paragraphs.	Gemini 2.5 Flash
Page setter	4 to 8 pages at whole sentences (Urdu ؟ ۔ and English marks), about 75 words a page, both languages split at the same points.	Plain code: deterministic, free, can't run out of quota
Art director	One calm painting brief per page, plus the cover.	Gemini 2.5 Flash-Lite (its own quota)
Illustrator	Paints every page in one fixed gouache style, three at a time, with a seed per family.	Cloudflare Workers AI · FLUX.1 schnell
Orchestrator	Hand-offs, fallbacks and a timed trace of every step.	Plain code

Routing has exactly one right answer at each step, so the orchestrator is deliberately not a language model: a model there would only add latency and another way to fail.

CHAPTER 06 / 10

How the AI stays honest

سچائی کی ضمانت

The prompts ask the models to behave. The code checks that they did.

Hudhud's guardrails

Never more than three clarifying questions, and never before the story is told. One question at a time. Questions asking for phone numbers, ID cards, bank details, money or addresses are rejected and retried. Cues of grief, 1947 violence or tiredness (in English, Roman Urdu and Urdu script) make it gentler or end the sitting.

The writer never invents

A pull quote not found word for word in a recording is dropped or repointed to the line that said it. A year never said (Western or Urdu digits) is dropped. Unclear parts are marked `[unclear]`, never guessed. Every page keeps the recordings it came from.

Safe pictures

The art director writes briefs with no faces of real people and nothing graphic, even for 1947. Graphic scenes are swapped for quiet symbols: a train, a lantern, a doorway. The painter only ever receives these briefs, never the elder's words or photo.

Prompt safety

Transcripts are fenced between `<<<` and `>>>` as data, so an answer can't act as an instruction. Models reply in JSON; every reply is validated in code, with one retry, before it is used.

Resilience: the book never stalls

- **Keys and models rotate.** When a Gemini model is out of quota, the backend tries every key, then the next free model (2.5 Flash → 2.5 Flash-Lite → 2.0 Flash).
- **Per-minute limits are waited out** (up to 35 seconds) for the book; a used-up daily quota is never waited for, and the live conversation never waits.
- **Every step has a fallback.** No edited story: the page setter lays out the elder's own words as a draft, still painted, with a "Write it properly" button for later. No briefs: the theme's calm scenes. A failed painting: a themed picture.
- **A pre-flight check**, `python -m app.check`, shows which key and model pairs still have quota today.

CHAPTER 07 / 10

Models, APIs and stack

ماڈل اور ٹیکنالوجی

Every third-party AI service, and what it does in the app.

SERVICE	MODEL	USED FOR
Google Gemini API	`gemini-2.5-flash`	Hudhud's decisions, the story writer
Google Gemini API	`gemini-2.5-flash-lite`	Art director's briefs, Urdu speech-to-text, fallback
Google Gemini API	`gemini-2.0-flash`	Second fallback when quotas run out
Google Gemini API	`gemini-2.5-flash-preview-tts`	Hudhud's spoken questions
Cloudflare Workers AI	`flux-1-schnell`	Page paintings
Cloudflare Workers AI	`whisper-large-v3-turbo`	Speech-to-text fallback
Supabase	(not AI)	Sign-in, Postgres, private storage

Architecture



Frontend

Next.js 16, React 19, TypeScript, Tailwind CSS 4, Motion. A hand-built 3D page-turn reader; PDF made in the browser with html-to-image and jsPDF so Nastaliq stays shaped.

Backend

Python 3.12, FastAPI, httpx, Pydantic; Docker on Render. Agents have no keys and no network: the API hands them a model and a painter, so tests use fakes. 70 automated tests.

CHAPTER 08 / 10

Setup and deployment

تنصیب

From a fresh clone to a running app.

Requirements: **Python 3.11+ with uv** and **Node 20.9+**. All services have free tiers; keys stay in git-ignored `.env` files.

```
# 1. Backend (the agents)
cd backend
uv init                     # first time only; generated files are git-ignored
uv venv
uv add -r requirements.txt
uv add --dev pytest
cp .env.example .env        # add keys; keep APP_ENV=development
uv run uvicorn app.main:app --env-file .env --reload --port 8000

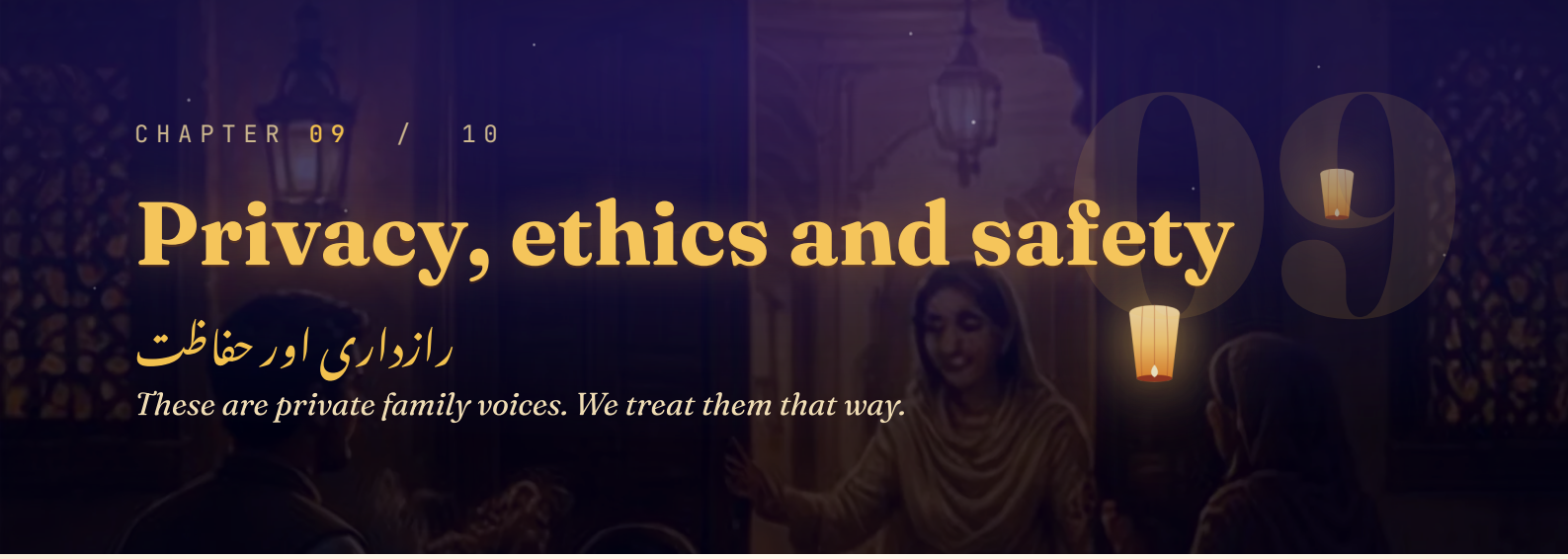
# 2. Frontend, from the repo root
npm install
cp frontend/.env.example frontend/.env.local   # NEXT_PUBLIC_API_URL=http://localhost:8000
npm run dev                                    # http://localhost:3000

# 3. Check every key and today's Gemini quota · run the tests
uv run python -m app.check
uv run pytest
```

KEY	WHERE	GOES IN
`GEMINI_API_KEY` / `GEMINI_API_KEYS`	Google AI Studio	`backend/.env`
`CLOUDFLARE_ACCOUNT_ID`, `CLOUDFLARE_API_TOKEN`	Cloudflare dashboard	`backend/.env`
Supabase URL and keys	Supabase project	both `.env` files

Deployment

Backend → **Render** (Docker, root `backend`, health check `/health`). **Frontend** → **Vercel** (root `frontend`, the four `NEXT_PUBLIC_` variables). Then add the Vercel URL to the backend's `ALLOWED_ORIGINS` and to Supabase's Auth URL settings. Database: run the five migrations in `backend/supabase/migrations` in order.



CHAPTER 09 / 10

Privacy, ethics and safety

رازداری اور حفاظت

These are private family voices. We treat them that way.

Consent first

The elder agrees on the record before anything is saved, and the consent clip is kept. Recording can stop at any moment.

Private by default

Row level security on every table; voices and paintings live in private storage served through short-lived signed links. Share links are read-only and unguessable (24 to 64 random characters).

No cloning, no inventing

Only the elder's real recordings are ever played; Hudhud's voice is the only synthetic one. The writer may use only what was said.

Keys stay on the server

Every AI and service key lives on the backend; the browser never sees one. AI calls are verified against the user's sign-in, rate-limited and size-capped.

Minimal data, no tracking

No ads, no analytics, no tracking cookies, and nothing is sold. Deleting a storybook removes its story, recordings and paintings.

Sensitive history, gently

1947 stories can carry grief. Hudhud recognises it and slows down; paintings never show violence, only quiet symbols.

Full details: the public **Privacy policy** (yaadon-ki-baithak.vercel.app/privacy) and **Terms of service** (yaadon-ki-baithak.vercel.app/terms).

CHAPTER 10 / 10

Limits, future and team

حدود، مستقبل اور ٹیم

What it can't do yet, where it goes next, and who built it.

Known limitations

- Free AI tiers have daily limits; when they run out, books are drafts in the elder's own words until quota returns.
- Transcription works best with clear audio; heavy background noise lowers accuracy.
- Urdu and English only for now; Hudhud's Urdu voice is a preview model.
- Paintings are illustrations, not historical records.
- The free backend sleeps when idle and takes up to a minute to wake.

What comes next

- Turn old WhatsApp voice notes into storybooks.
- Punjabi, Sindhi, Pashto and Saraiki.
- A printed hardcover ordered from the app.
- A family tree linking stories, people and places.

Team contributions

MEMBER	ROLE	WHAT THEY BUILT
Ayesha Mughal	Team lead · frontend · UI/UX · AI agents	The interface and design system (home, recording room, storybook reader, bookshelf, PDF, sharing) and the agents: Hudhud, story writer, page setter, art director, illustrator, orchestrator.
Faria	Backend developer	The FastAPI backend: routes, authentication, rate limits, provider clients and storage.
Humema Israr	Documentation · presenter	Documentation, the demo video and the presentation.

Development tools: AI coding assistants were used during development; all code was reviewed and tested by the team.
Acknowledgements: Next.js, React, Tailwind CSS, Motion, FastAPI, Pydantic, httpx, Supabase, lucide-react, html-to-image, jsPDF, Google Gemini, Cloudflare Workers AI, and the Fraunces, Nunito and Noto Nastaliq Urdu typefaces.



یادوں کی بیٹھک

Before the stories are gone.

yaadon-ki-baithak.vercel.app

github.com/CodeVoyager007/Yaadon-Ki-Baithak

API · yaadon-ki-baithak.onrender.com

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